

Morgan Bleasdale

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Education

Imperial College London

September 2025 – Present

MEng Electrical and Electronic Engineering

On track for First-Class Honours (graduation 2029)

Kings Monkton School, Cardiff

A Levels: Maths (A), Further Maths (A*), Physics (A*), Computer Science (A), EPQ (A*)* September 2023 – July 2025

*GCSEs: 11 A*s, 1 A and 1 Distinction (Including Maths and English)* September 2021 – July 2023

Awards: Principal's Prize for Academic Achievement, KS5 Academic Achievement Award, Jean Morris Prize for Excellence in Mathematics, Pupil Leadership Prize, McCarthy Prize for Business

Experience

Vishay Intertechnology, Inc | Newport, UK

July 2026 – Present

Semiconductor Engineering Summer Internship

- Diagnosed incorrect Active Area Deposition control limit assignment across 190 MOSFET parts, by identifying critical parameters; recalculated and reassigned limits to create a data-driven recipe-selection framework.
- Implemented upper specification limits to end-of-line BVDSS testing to prevent out-of-spec MOSFETs being shipped undetected, using 8D methodology; presented to Process Change Board for integration in facility.
- Investigated a silicon staining defect affecting ~3% of production wafers by analysing focused ion beam (FIB) failure analysis images and comparing to fab tool data to identify potential root causes.
- Updated documentation describing silicon carbide (SiC) diode manufacturing process.

UKESF Semiconductor Talent Award | London, UK

September 2025 – Present

Talent Award Recipient and Mentee

- Nationally recognised for academic excellence and future potential in semiconductor engineering, awarded a competitive, merit-based scholarship (£11,000 in first 2 years, with continued funding in subsequent years).
- Attending mentorship sessions from UK semiconductor figures, including the IET and Silicon Catalyst Europe.

National Aeronautics and Space Administration (NASA) | Florida, USA

July 2024 – August 2025

VP of Engineering, International Space Settlement Design Competition

- Selected twice to represent the UK (top 12 of 1000+ applicants), leading team of 60+ engineering students to design a space station. **2024: Winner** (Lunar data centre), **2025: Runner-up** (Venusian Mining Settlement).
- Led overall technical design, managing 4 engineering departments with the systems engineering V-Model and a requirements traceability matrix to ensure 50-slide proposal was cohesive and complied with RFP.
- Directed Q&A section of presentation in front of panel of NASA and SpaceX engineers, and 250+ spectators.

Engineering Education Scheme Wales | Cardiff, UK

October 2023 – December 2024

Project Coordinator and Award Winner

- Designed a sustainable housing estate for 500+ residents, with a grid-tied hybrid power microgrid, intelligent EV charging infrastructure, and decentralised geothermal heating system with ground source heat pumps.
- Led team of 6 using an iterative Agile approach to present 73-page technical report to industry professionals.
- Awarded Best Use of STEM for Sustainability by Welsh Government and named runner-up as **Welsh Engineering Student of the Year 2024**; achieved Gold CREST award for engineering project excellence.

Recent Projects

RISC CPU in SystemVerilog with custom ISA

June 2026 – July 2026

16-bit Harvard-architecture RISC CPU in SystemVerilog, following the design principles of ARM architecture.

- Implemented a single-cycle data path with an 8-operation ALU and NZCV flag registers, where flag states are updated in one cycle and consumed by the next cycle to support carry-chained operations and jumps.
- Flag-based conditional jumping (16 conditions, derived from NZCV status flags set by ALU operations), enabling structured control flow such as loops, conditionals, and subroutine calls/returns.
- Built load/store architecture with dedicated memory instructions, interfacing an 8x16-bit register file with data RAM and PC register with instruction ROM, supporting register-offset or direct addressing.
- Verified RTL using SystemVerilog testbenches in Icarus Verilog with a 100% pass-rate; demonstrated functionality with a custom assembly program to perform a multi-word 16x16-bit software multiplication.

Multi-sensor remote-controlled Wi-Fi Lunar Rover

May 2026 – June 2026

Built an RC rover that surveys a lunar environment, detecting RF, IR, ultrasonic signals, and magnetic fields.

- Built radio subsystem with a tuned LC coil antenna, inverting amplifier, envelope detector, and hysteresis comparator to demodulate an 89 kHz ASK-modulated carrier into binary UART-framed ASCII data.
- Implemented differential-drive rover movement via PWM speed control and direction; integrated with an HTTP server/webpage over Wi-Fi for remote driving and live telemetry.
- Led project management (6 people) using Agile sprints and authored the 62-page technical report.

Discrete BJT Operational Amplifier (LTSpice)

February 2026 – March 2026

Designed an operational amplifier from individual BJTs, following a differential-gain-power stage architecture.

- Designed a differential amplifier (long-tailed pair) with a Wilson current mirror active load for high differential gain, low common mode gain, and a high output impedance for improved CMRR.
- Implemented a common-emitter amplifier stage with current mirror active load to boost voltage gain.
- Developed class AB push-pull power amplifier to increase output current and reduce crossover distortion.
- Diagnosed high-frequency instability via AC analysis of phase response (negative phase margin); stabilised with a compensation capacitor to bypass common-emitter stage at high frequencies.

Skills

Digital Design & Architecture

Verilog/SystemVerilog, RTL Design, FPGA (Intel Quartus), RISC CPU Architecture, Verification (Icarus Verilog)

Analogue & Embedded Systems

Operational Amplifiers, Passive/Active Filters, RF systems, PCB Design (KiCad), Arduino, Raspberry Pi

Software

C++, C#, Python, JavaScript, TypeScript, React (Next.js, Vue.js), PHP, Git, SQL, Supabase, Vercel, OAuth

Volunteering

UK Space Design Competition

October 2025 – Present

- Provided technical advice and pastoral support to groups of 40-60 student competitors (Years 10-13) as they worked in groups to create and present engineering proposals over 1 or 2-day events.

Kings Monkton School, Head Student

September 2024 – July 2025

- Raised £1,000+ through charity events and initiatives, including managing partnerships with organisations.
- Organised school-wide community and cultural events, aimed at engaging students with real-world problems beyond academic study, such as a climate debate (70+ participants).